

Franklin Muoghalu

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EDUCATION

University of Houston

May 2026

Bachelor's in Computer Science

- GPA of **3.57/4.0**; Dean's list student

TECHNICAL SKILLS

Programming Languages: Python, C++, C, C#, Java, JavaScript, SQL, R, HTML/CSS, Razor

Frameworks & Libraries: React.js, Node.js, Express.js, Flask, .NET

Tools & Platforms: Git, GitHub, GitHub Actions, Docker, Google Cloud, Mysql, Postgres

Server & DevOps: Nginx, Reverse Proxy Configuration, SSL/TLS, Cloudflare DNS, SSH, Systemd Services

AI & ML: RAG, NLP, Model Fine-Tuning, Neural Networks, Logistic Regression

WORK EXPERIENCE

Software Engineer, Foxconn

May 2026– Present

- Architected an asynchronous PostgreSQL database configuration listener utilizing Python's arq worker and database triggers, enabling real-time background updates to system cron schedules
- Designed and deployed multi-tier architectures and microservices using Docker Compose and Nginx reverse proxies to manage isolated internal container communication and route public web traffic
- Developed and maintained multiple backend routing configurations and database schemas across different teams using FastAPI, Flask, and ASP.NET Core

Tool & System Engineer, Foxconn

May 2025 – May 2026

- Designed and deployed internal testing software using Python and C#, automating manual processes to increase overall testing efficiency by 30%
- Engineered a suite of 5 custom full-stack applications with React to resolve critical operational bottlenecks across different departments
- Authored comprehensive technical documentation and system guidelines, accelerating technician workflows and reducing new-hire onboarding time by 20%

PROJECT EXPERIENCE

Self-Hosted Server Infrastructure & Deployment | Nginx, Docker, Cloudflare

October 2024 – Present

- Built and maintained a multi-platform server infrastructure spanning a Raspberry Pi 5, a home Linux server, and an Oracle Cloud instance
- Hosted and managed multiple applications including personal web apps, LLM tools, and a multiplayer game server, ensuring high availability and secure access
- Configured a full Nginx-based reverse proxy with Cloudflare SSL/TLS, DNS, and firewall protections across all environments
- Deployed database services on Oracle Cloud, supporting API backends and real-time application features through optimized connection management
- Implemented automated updates, containerized services, and monitoring tools to streamline maintenance and system reliability

Bone Pile Management System | Python, FastAPI, React, Docker, Arq, PostgreSQL, Redis

March 2026 – May 2026

- Built a tool for the management and organization of board data within the bone pile
- Implemented a backend API using FastAPI to handle CRUD operations for bone pile data, ensuring efficient data management and retrieval
- Implemented a RAG model to provide engineers with relevant data as needed
- Boosted overall efficiency of the failure analysis department by 25%

Text to Website | Python, Httpx, Flask, React, LLM, MCP

September 2025

- Developed a custom MCP server-client system that automatically generates frontend design skeletons to accelerate UI development
- Increased overall web development efficiency by 40% through automated component generation and streamlined workflows
- Integrated LLM-driven prompts and template generation to produce consistent, responsive layouts with minimal effort

O.M.N.I | Python, LLMs, Embeddings, NLP, RAG, OpenAI, Docker, Cloudflare

November 2024 - May 2025

- Won the Grand Prize at the Johnson Space Center Hackathon for applying O.M.N.I to NASA mission manuals, enabling astronauts to rapidly retrieve emergency procedures using natural-language queries
- Built an intelligent AI agent with a Python backend and Tkinter UI, designed to allow users to query and navigate complex technical information in real time
- Developed a publicly accessible API capable of ingesting documents from any source, generating embeddings, and storing them in a vector database for high-accuracy retrieval
- Containerized and deployed the system using Docker and Cloudflare Tunnels for secure, remote accessibility and testing

Simulating Clocktimes | C++

September 2024

- Simulated CPU and user process scheduling using a priority queue, differentiating between high and low priority events.
- Implemented time-sharing logic to ensure equitable CPU allocation among processes, improving system responsiveness by 20%.
- Utilized C++ to build the simulation, incorporating process management, queues, and event handling.
- Tested and validated the simulation with various input scenarios to ensure accurate and reliable performance.